Discovery Lab

Exam 4 Revision

Lower Secondary Stage 2 · Science · Lessons 17 to 20

Acids and bases, electric circuits, series and parallel circuits, and changing the current.



Lesson 17: Acids and Bases

Lesson 18: Electric Circuits

Lesson 19: Series and Parallel Circuits

Lesson 20: Changing the Current

Name: _____ Class: _____ Date: _____

LESSON 17

17 Acids and Bases

What acids and bases are, and what happens when they meet.

Key ideas

Acids have a **pH less than 7** (vinegar, lemon juice, hydrochloric acid).

Bases have a **pH greater than 7**. A base that dissolves in water is an **alkali** (toothpaste, baking soda, sodium hydroxide).

Neutralisation: acid + base → **salt + water**. Acid + carbonate → salt + water + **carbon dioxide**. Acid + metal → salt + **hydrogen**.

Antacid tablets are bases that neutralise extra stomach acid. Lime neutralises acidic soil.

Indicators: litmus is red in acid and blue in alkali; phenolphthalein is pink in alkali; methyl orange is red in acid and yellow in alkali.

Acid + base	Salt formed
Hydrochloric acid + sodium hydroxide	Sodium chloride
Sulfuric acid + potassium hydroxide	Potassium sulfate
Nitric acid + calcium hydroxide	Calcium nitrate
Hydrochloric acid + magnesium oxide	Magnesium chloride

DID YOU KNOW?

Antacid tablets for heartburn are neutralisation happening inside your body!

Check yourself

- [ ] What pH do acids and alkalis have?
- [ ] What is the word equation for neutralisation?
- [ ] Which gas makes vinegar and baking soda fizz?
- [ ] Name the salt made by hydrochloric acid and sodium hydroxide.
- [ ] Why are antacid tablets useful?

REVISION TASK

Complete the salt table from memory, then colour the pH scale from red (acid) to purple (alkali).

LESSON 18

18 Electric Circuits

A complete path for electricity to flow.

Key ideas

A circuit needs a **power source** (cell or battery), **wires** and a **component** (bulb, buzzer, motor).

If the circuit is **broken**, no current flows. A **switch** opens or closes it.

A battery stores **chemical energy** and changes it to electrical energy.

Wires are **metal** because metals are good **conductors**; plastic is an **insulator**.

An **ammeter** measures current (in series). A **voltmeter** measures voltage (across a component). A **fuse** melts if the current is too big.

Component	Function
Battery	Powers the circuit
Wire	Allows current to flow
Bulb	Lights up when current flows
Buzzer	Makes sound when current flows
Switch	Opens or closes the circuit

DID YOU KNOW?

Thomas Edison's first practical bulb in 1879 lasted over 13 hours; later ones lasted 1,200 hours.

Check yourself

- [ ] What are the three parts needed to complete a circuit?
- [ ] What happens if the circuit is broken?
- [ ] What does a switch do?
- [ ] What type of energy is in a battery?
- [ ] Why are metals used for wires?

REVISION TASK

Draw a circuit diagram with a cell, a switch and a lamp, using the correct symbols.

LESSON 19

19 Series and Parallel Circuits

One loop or many branches: how the layout changes everything.

Key ideas

Series: one loop. Current is the **same** everywhere. Voltage is **shared**. If one bulb goes out, **all** go out.

Parallel: branches. Current is **shared** across branches. Each branch gets the **full** voltage. If one bulb goes out, the others **stay on**.

More bulbs in series make them **dimmer**. More bulbs in parallel do not dim the others.

Household lights are wired in **parallel** so each one works on its own.

Feature	Series circuit	Parallel circuit
Number of loops	One	More than one
If one bulb fails	All bulbs go off	Others stay on
Current flow	Same all around	Shared between branches
Brightness of bulbs	Dimmer with more bulbs	Stay bright

DID YOU KNOW?

Old fairy lights were wired in series: if one bulb blew, they all went out.

Check yourself

- [ ] What is the difference between a series and a parallel circuit?
- [ ] What happens if a bulb breaks in a series circuit?
- [ ] Why are household lights wired in parallel?
- [ ] Which circuit type shares current across branches?
- [ ] Which type keeps other bulbs on if one goes out?

REVISION TASK

Draw a series and a parallel circuit with three bulbs each and label what happens when one bulb is removed.

LESSON 20

20 Changing the Current

Voltage, resistance and current, and how they are linked.

Key ideas

Current (amps, A) is the flow of electrons. **Voltage** (volts, V) is the push. **Resistance** (ohms, Ω) slows the current.

V = I × R, so I = V ÷ R and R = V ÷ I.

More voltage = **more** current. More resistance = **less** current.

In series, resistance **adds up**. In parallel, the current has more paths so resistance falls.

Longer wires have more resistance; thicker wires have less.

Setup	Current	Why?
1 battery, 1 bulb	Medium	Low voltage, low resistance
2 batteries, 1 bulb	High	More voltage = more current
1 battery, 2 bulbs in series	Low	More resistance = less current
1 battery, 2 bulbs in parallel	Medium in each branch	Each branch has the full voltage

DID YOU KNOW?

Lightning can reach 30,000 amps, compared to about 1 amp in a small torch.

Check yourself

- [ ] What is electric current?
- [ ] How does resistance affect current?
- [ ] What unit measures current?
- [ ] What happens when you add more bulbs to a series circuit?
- [ ] A 12 V supply drives 4 A. What is the resistance?

REVISION TASK

Use V = I × R to work out three missing values, then predict the current in each setup in the table.

FINAL PRACTICE

Final practice

Tick each box when you can answer without looking.

Final practice

- [ ] What is a neutralisation reaction?
- [ ] Name 3 components of a working electric circuit.
- [ ] What are the advantages of parallel circuits in homes?
- [ ] What happens when you increase voltage in a circuit?
- [ ] How does resistance affect the brightness of bulbs?

CHALLENGE

Explain how a change in resistance or voltage affects the current in both series and parallel circuits.

Revision checklist

Revision task	Done?
Reviewed the Canvas slides for Lessons 17–20	☐
Answered every "Check yourself" question	☐
Practised the matching and table activities	☐
Used the 3D labs to compare series and parallel	☐
Practised short answers and the practice exam	☐
Reviewed the keywords: current, voltage, resistance, neutralisation	☐

Try it live · Practise with 3D labs, games and a practice exam at **morehumaned.com/discovery-lab**

ANSWER KEY

Answer key

Check your own answers against these. If one is hard, go back to the lesson.

Lesson 17: Acids and Bases

1. Acids are below 7 and alkalis are above 7.
2. Acid + base → salt + water.
3. Carbon dioxide.
4. Sodium chloride.
5. They contain bases that neutralise extra stomach acid.

Lesson 18: Electric Circuits

1. A power source, wires and a component.
2. Electricity cannot flow, so the circuit stops working.
3. It opens or closes the circuit.
4. Chemical energy.
5. Metals are good conductors.

Lesson 19: Series and Parallel Circuits

1. Series has one loop; parallel has branches.
2. All the bulbs go out.
3. So each light works on its own and stays bright.
4. Parallel.
5. Parallel.

Lesson 20: Changing the Current

1. The flow of electric charge (electrons).
2. More resistance means less current.
3. The ampere (amp, A).
4. Resistance increases, so the current falls and the bulbs get dimmer.
5. R = V ÷ I = 12 ÷ 4 = 3 Ω.

Final practice

1. An acid reacts with a base to make a salt and water.
2. For example a cell, wires and a bulb (or a switch, buzzer).
3. Each appliance works on its own and keeps full brightness.
4. The current increases.
5. More resistance means less current, so the bulbs are dimmer.

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You are going to do great! Use this booklet to plan your study time, then bring it all together.

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